A Short Proof of a Range-Relative FPTAS for Polynomial Integer Optimization over Bounded Polyhedra

Robert Hildebrand*

October 14, 2026

Abstract

We give a short proof of a fully polynomial-time range-relative approximation scheme for polynomial optimization over the integer points of a bounded rational polyhedron in fixed dimension. Boundedness supplies a finite slack decomposition and a finite integer objective range. The proof combines the reflecting-set decomposition of Cook, Hartmann, Kannan, and McDiarmid with Markov's polynomial derivative inequality and uses one integer linear feasibility query per refined cell. The running time is polynomial in the input encoding size, the numerical degree, and the reciprocal accuracy.

1 Introduction and main result

Let $P = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{b}\}$ be a bounded rational polyhedron, with $A \in \mathbb{Q}^{m\times d}$ and $\mathbf{b} \in \mathbb{Q}^m$, and let $f \in \mathbb{Z}[x_1, \ldots, x_d]$. We study

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b}, \\ & \mathbf{x} \in \mathbb{Z}^d. \end{aligned} \tag{1}$$

When $P \cap \mathbb{Z}^d \neq \varnothing$, choose minimizers and maximizers $\mathbf{x}_{\min}, \mathbf{x}_{\max}$ and write $f_{\min} = f(\mathbf{x}_{\min})$ and $f_{\max} = f(\mathbf{x}_{\max})$.

We use d for dimension, D for total degree, and $\mathbf{x}_\varepsilon$ for the returned point, following [1]. The range-relative guarantee is their weak approximation guarantee, stated here for minimization. It is invariant under translation of the objective. It is different from a multiplicative approximation to a nonnegative minimum.

An interactive explanation, supporting material, and partial Lean verification are available in the project repository.[1]

De Loera, Hemmecke, Köppe, and Weismantel established fully polynomial approximation schemes based on short rational generating functions, including nonnegative polynomial maximization and range-relative approximation of arbitrary polynomial objectives in fixed dimension [2, 1]. Del Pia gives a geometric range-relative approximation algorithm for bounded mixed-integer quadratic programs with fixed quadratic rank and fixed number of integer variables [3]. We give an alternative construction for the bounded, pure-integer problem. The purpose is to obtain a short proof using linear feasibility, with explicit dependence on the encoding length.

*Developed with assistance from OpenAI's Codex assistant; see the detailed AI-assistance and contribution disclosure at the end of the paper.

[1] `https://github.com/RobertHildebrand/FPTAS-Polynomial-Optimization`

The common geometric ingredient is a decomposition by relative slack scales. The dyadic reflecting sets of Cook–Hartmann–Kannan–McDiarmid (CHKM) are central to their integer-hull analysis [4]. Symmetric displacement covers likewise underpin the quadratic and cubic algorithms of Ari and Hildebrand [5]. In parallel work on exact integer quadratic programming, Wei explicitly uses dyadic grouping of constraint slacks to obtain regions supporting two-sided feasible moves [6, Section 1.1]. These uses suggest that relative-slack decompositions are a useful common tool for polynomial integer optimization: they convert proximity measured against the constraints into feasible algebraic comparisons, without reference to objective coefficients.

Here refinement strengthens one-step reflection to a long feasible arithmetic progression. Restricting a degree-D polynomial to that progression and applying Markov's polynomial derivative inequality [7, Theorem 5.1.8] then yields an objective-independent representative set that approximates both integer extrema; see Proposition 2.7. Thus the geometric cover serves all polynomial objectives of the prescribed degree simultaneously. The essential feature is multiplicative control of positive slacks, not the choice of base two: Remark 2.3 makes the connection to extrapolation exact. Piazzon and Vianello [8, Propositions 1–2] obtain range-relative approximation from norming meshes by choosing arbitrary feasible representatives in sufficiently small boxes. Our refined reflecting cells instead supply feasible integer progressions, so the error is measured against the integer objective range.

Markov's inequality is essential to the polynomial cell bound: it controls variation relative to the integer objective range, so the required refinement depends only on degree and accuracy. A coefficient-based Lipschitz estimate alone could require refinement exponential in the input encoding size. The Lipschitz-based grid reduction of [1] can tolerate a large denominator because its integer solver depends polynomially on the denominator's bit length; explicit cell enumeration requires polynomially many cells.

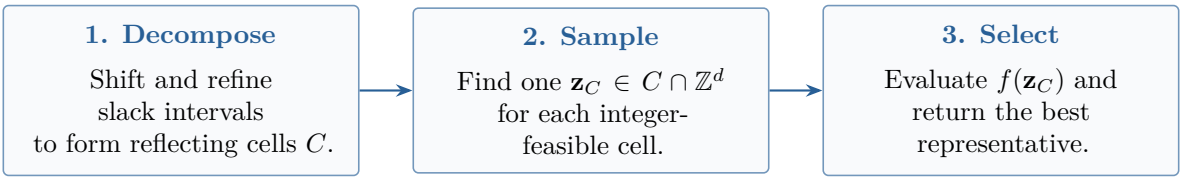


Figure 1: Algorithm at a glance: decompose a relaxation with the same integer points, select one representative from each cell containing integer points, and return the representative with the best objective value.

Figure 1 summarizes the algorithm. No local approximation to the objective is optimized. Instead, reflection controls the error at a global optimum, which is enough to justify the final selection. Section 2 proves this estimate, Section 3 gives the algorithm and its bit complexity, and Section 4 discusses extensions and related decomposition methods.

More broadly, the method follows a decompose–solve–select pattern: a local error bound on the cell containing an optimum transfers to the best returned candidate. For polynomial optimization, this permits other cell decompositions or tractable local models when suitable error estimates are available.

The decomposition also connects to the geometric subdivision of Papadimitriou and Yannakakis [9] and its use for affine slicing in Hildebrand, Weismantel, and Zemmer [10, Section 2]. Applying multiplicative bins to the slacks $s_i(\mathbf{x}) = b_i - \mathbf{a}_i^\top \mathbf{x}$ gives the same relative-width property as our refined dyadic intervals: a bin $[a, (1+\delta)a]$ has width at most δa. Points in a common block therefore extend to feasible progressions of length $1/\delta$, with zero slacks handled separately. As in the

proof of [10, Theorem 2.1], the bins are pulled back to an arrangement in the original variable space $\mathbb{R}^d$; enumerating all combinations in the m-dimensional slack space would lose the fixed-dimensional complexity bound. Unlike the sliceable-objective framework of [10], our approximation argument uses Markov's inequality and measures error relative to the objective range.

To state the complexity bound precisely, we fix the following encoding convention. Vectors are columns and are written in bold. For an integer u, its *binary encoding size* is $\langle u \rangle = 1 + \lceil \log_2(1+|u|) \rceil$, counting a sign bit and the binary magnitude bits. We use *encoding size*, *bit length*, or simply *size* for storage measured in bits when the context is clear; *magnitude* refers to the numerical absolute value. A rational number u/v is stored in lowest terms with $v > 0$, with size $\langle u/v \rangle = \langle u \rangle + \langle v \rangle$. For matrices, vectors, and lists, $\langle \cdot \rangle$ is the sum of the entry sizes together with the binary encoding sizes of the dimensions and list lengths. The polynomial is supplied explicitly as

$$f(\mathbf{x}) = \sum_{\boldsymbol{\alpha} \in \mathcal{A}} c_{\boldsymbol{\alpha}} \mathbf{x}^{\boldsymbol{\alpha}}, \qquad c_{\boldsymbol{\alpha}} \in \mathbb{Z}, \quad \boldsymbol{\alpha} \in \mathbb{Z}_+^d, \quad |\boldsymbol{\alpha}|_1 \le D.$$

Following [1], coefficients are encoded in binary and exponent vectors in unary: each exponent α_j is stored as α_j one-bits followed by a zero, using $\alpha_j + 1$ bits. The input is therefore a bit string, although exponents use unary rather than positional binary notation. Accordingly, $\langle f \rangle$ includes the coefficient sizes and $\sum_{\boldsymbol{\alpha} \in \mathcal{A}} \sum_{j=1}^{d} (\alpha_j + 1)$, together with dimensions and list lengths. Write $\varphi = \langle A, \mathbf{b}, f, \varepsilon \rangle$ for the total input encoding size. We set $D = \max\{1, \deg f\}$ from the nonzero monomials, taking $D = 1$ for the zero polynomial; no separate degree input is needed. We retain D explicitly in the running-time bound to show its contribution.

Definition 1.1 (Range-relative FPTAS)**.** Under the encoding convention above and for fixed dimension d, a range-relative fully polynomial-time approximation scheme is an algorithm that, for every rational $0 < \varepsilon \le 1$, reports infeasibility when $P \cap \mathbb{Z}^d = \varnothing$, and otherwise returns $\mathbf{x}_\varepsilon \in P \cap \mathbb{Z}^d$ satisfying

$$0 \le f(\mathbf{x}_\varepsilon) - f_{\min} \le \varepsilon (f_{\max} - f_{\min}). \tag{2}$$

Its running time is polynomial in the total input encoding size φ and $1/\varepsilon$. The polynomial may depend on the fixed dimension. This is the fully polynomial-time weak-approximation notion of [1]. The maximization definition uses $f_{\max} - f(\mathbf{x}_\varepsilon)$ on the left.

Running times are measured in bit operations; oracle calls and arithmetic operations are counted separately where stated. The function poly denotes a polynomial with an absolute exponent, independent of d, D, ε; different occurrences may denote different polynomials. Logs are base two unless indicated otherwise.

Theorem 1.1 (Polynomial approximation by reflecting cells)**.** *Let $d \ge 1$ and let $P = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \le \mathbf{b}\}$ be bounded, with $A \in \mathbb{Z}^{m \times d}$ and $\mathbf{b} \in \mathbb{Z}^m$, let $f \in \mathbb{Z}[x_1, \dots, x_d]$, and put $D = \max\{1, \deg f\}$, with $D = 1$ when $f \equiv 0$. For rational $0 < \varepsilon \le 1$, let $\varphi = \langle A, \mathbf{b}, f, \varepsilon \rangle$ be the total input encoding size. There is a deterministic algorithm that reports infeasibility or returns $\mathbf{x}_\varepsilon \in P \cap \mathbb{Z}^d$ satisfying* (2)*, in at most*

$$\left(\frac{\varphi D}{\varepsilon} \right)^{O(d)} \tag{3}$$

bit operations, with an absolute constant in the $O(d)$ exponent.

Since the input explicitly lists the m inequalities in d variables, $m, d \le \varphi$; their separate factors in the detailed running-time bound are therefore absorbed into the base φ and the $O(d)$ exponent in (3). The dependence on dimension remains explicit in that exponent. More precisely, for fixed

P, D, d, the algorithm makes $O(\varepsilon^{-d/2})$ integer-feasibility queries; the detailed running-time bound in the proof retains the factor $\max\{D^2, D/\sqrt{\varepsilon}\}$.

Rational inequalities can be normalized to integral data in polynomial time, as detailed below. Rational objective coefficients can likewise be reduced to integer coefficients by clearing denominators in polynomial time; positive scaling preserves the approximation guarantee, as in [1, Section 5]. The remaining construction parameters and an explicit upper bound on the number of feasibility queries are introduced in the proof. For fixed d, the bound is polynomial in φ and $1/\varepsilon$, hence gives an FPTAS in Definition 1.1. The same explicit bound remains valid for binary-encoded exponents, but then its numerical dependence on D does not assert an FPTAS in the sparse binary input size.

The following nonnegative-maximization guarantee was established by De Loera et al. [2, Theorem 1.1]; it follows from our range-relative bound by the same elementary conversion.

Corollary 1.2 (Nonnegative maximization [2, Theorem 1.1])**.** *Under the same assumptions and encoding conventions, if $f \geq 0$ on $P \cap \mathbb{Z}^d$, choosing the largest objective value among the representatives yields*

$$f(\mathbf{x}_\varepsilon) \geq (1-\varepsilon) f_{\max}.$$

Thus the construction gives a multiplicative FPTAS for nonnegative maximization under the encoding convention of Definition 1.1.

For the mixed-integer result, we follow the integral-slice scaling, barycentric rounding, and reduction to integer optimization of De Loera et al. [1, Section 3], who established the approximation guarantees in [1, Theorems 1(a) and 2(b)]. We improve the grid multiplier in this reduction by using Markov's inequality in place of their coefficient-based Lipschitz estimate. Combining the resulting coefficient-independent multiplier with Theorem 1.1 yields the explicit running-time bound below.

Corollary 1.3 (Mixed-integer approximation)**.** *Fix the total dimension $d = d_c + d_i$. Let $P = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{b}\}$ be a bounded rational polyhedron and $f \in \mathbb{Z}[x_1, \ldots, x_d]$. Under the same encoding convention, for rational $0 < \varepsilon \leq 1$ there is an algorithm, polynomial in $\varphi = \langle A, \mathbf{b}, f, \varepsilon \rangle$ and $1/\varepsilon$, that reports infeasibility or returns*

$$\mathbf{x}_\varepsilon \in P \cap (\mathbb{Q}^{d_c} \times \mathbb{Z}^{d_i}), \qquad 0 \leq f(\mathbf{x}_\varepsilon) - f_{\min} \leq \varepsilon (f_{\max} - f_{\min}),$$

where the extrema are over $P \cap (\mathbb{R}^{d_c} \times \mathbb{Z}^{d_i})$. For nonnegative f on this set, a multiplicative FPTAS for maximization is obtained as well. The running time is at most

$$\left(\frac{\varphi D}{\varepsilon}\right)^{O(d(d_c+1))} \tag{4}$$

bit operations, with an absolute constant in the exponent.

Comparison with prior algorithms The rational generating functions in these algorithms build on Barvinok's lattice-point counting method [11]. The guarantee in (2) is already provided by De Loera, Hemmecke, Köppe, and Weismantel [2, 1], whose generating-function algorithms run in time polynomial in the input length, numerical degree, and $1/\varepsilon$ for fixed total dimension. Their integer nonnegative-maximization subroutine computes weighted lattice sums for f^q, with $q = O(1 + \varepsilon^{-1} \log N)$ and $N = |P \cap \mathbb{Z}^d|$; solution recovery uses further weighted sums on restricted polytopes [2, Section 3]. Thus the work includes both Barvinok cone decomposition and subsequent symbolic calculations. A standard bound for the underlying lattice-point counting algorithm is $\langle A, \mathbf{b} \rangle^{O(d \log(d+1))}$ [12, Theorem 2.5] (under their arithmetic-cost convention; see Section 8.8),

whereas (3) gives $\varphi^{O(d)}$ for our entire algorithm when D and ε are fixed. This compares available upper bounds, not a lower bound on their algorithm; the counting bound alone does not describe their full FPTAS. Their published analysis does not make all dimension-dependent exponents in the weighted-summation cost explicit. Our construction instead gives the explicit bound (3) through cell enumeration, integer linear feasibility, and exact polynomial evaluation, and produces a representative set independent of the objective coefficients. These differences do not establish practical superiority: our cell count can be large, while generating-function methods can benefit from small cone decompositions.

Del Pia's construction shares the partition-and-solve structure, but uses different geometry [3, Sections 3–6]. After transforming the problem to spherical form, the aligned-vector case partitions the k quadratic coordinates into uniform boxes and minimizes an affine underestimator on each box by MILP; the remaining cases use flatness-based recursion. It does not use geometric slack bins. Our representatives instead require only feasibility because the reflected sample at an integer minimizer controls the first-order contribution.

For fixed quadratic rank k and integer dimension p, Del Pia [3, Proposition 3 and Section 6.2] uses $O_{k,p}(1+\varepsilon^{-k/2})$ mixed-integer linear optimization subproblems, with polynomial bit overhead for fixed k, p, while allowing variable continuous dimension. For our pure-integer algorithm, with fixed P, D, d, the counts proved in Section 3 are

$$O(\varepsilon^{-d/2}) \text{ integer feasibility calls}, \qquad O(\varepsilon^{-(d+1)/2}) \text{ enumeration LP tests}.$$

The per-test arithmetic has polynomial dependence on the encoding length $\langle\varepsilon\rangle$. The feasibility-call exponent matches $k/2$ when $k = d$, but the enumeration overhead remains and the rank-sensitive bound for $k < d$ is not recovered. Our construction instead applies to arbitrary degree and supplies one objective-independent representative set. Its dimension-dependent factor includes both the constraint normals and right-hand sides through their binary encoding lengths; the displayed bound is not a fixed-parameter tractability claim in d.

2 Reflecting cells and approximation

The proof separates the geometry of the feasible set from the behavior of the objective. We first partition the integer points into cells whose pairs extend to long feasible progressions. We then restrict the polynomial to such a progression through a global minimizer. The reflected point rules out a large first-order improvement in the backward direction, and a second-difference estimate controls the error of the forward representative. Neither the minimizer nor the objective range needs to be known by the algorithm.

The construction begins by making the slack gap explicit. If the input inequalities are rational, multiply each row by the product of the positive denominators in that row. This gives integral $A, \mathbf{b}$ in polynomial time, with total encoding size polynomial in the original input size φ. Each normalized entry has bit length $O(\varphi)$, since a row product has bit length at most the sum of its denominator bit lengths. The detailed bounds below refer to these normalized data; substituting their polynomial encoding bounds gives a bound in the original input length. No coefficient magnitude appears as a polynomial-time parameter.

Explicit equalities can be represented by pairs of inequalities. Lower-dimensional polytopes are allowed; no full-dimensionality assumption or integer change of coordinates is needed. Continuous emptiness and boundedness are checkable by rational linear programming. Boundedness is a hypothesis of the approximation theorem; an unbounded polyhedron is outside its scope.

For the remainder of the construction, $A, \mathbf{b}$ denote the normalized integral data; φ continues to denote the original input size. To track the construction explicitly, define

$$\tau = \lceil \log_2(2 + \max\{\|A\|_{\max}, \|\mathbf{b}\|_\infty\}) \rceil . \tag{5}$$

Here $\|A\|_{\max}$ is the largest absolute matrix entry, so τ bounds the coefficient bit lengths up to a constant. The remaining parameters have the following roles: Φ gives the common magnitude bound 2^Φ for coordinates and shifted slacks and determines how many intervals $[2^k, 2^{k+1}]$ are needed below, ℓ is the refinement, and H bounds the number of hyperplanes. Set

$$\Phi = (d+1)(\tau + \lceil \log_2(d+1) \rceil) + 4, \tag{6}$$

$$\ell = \max\{D^2, \lceil D/\sqrt{\varepsilon} \rceil\}, \qquad H = m((\Phi+2)\ell + 1), \tag{7}$$

$$R_d(H) = \sum_{j=0}^{d} \binom{H}{j} \le (1+H)^d. \tag{8}$$

Following the outward perturbation in [4, proof of Theorem 2.1, p. 29], define

$$\widetilde{P} = \{\mathbf{x} : A\mathbf{x} \le \mathbf{b} + \tfrac{1}{2}\mathbf{1}\},$$
$$s_i(\mathbf{x}) = b_i + \tfrac{1}{2} - \mathbf{a}_i^\top \mathbf{x}.$$

Since $A, \mathbf{b}$ are integral, $\widetilde{P} \cap \mathbb{Z}^d = P \cap \mathbb{Z}^d$. Every integer point of P has $s_i \ge 1/2$. We will extrapolate real segments in $\widetilde{P}$ and use integrality to return their integer samples to P. This shift is applied only to the pure-integer problem, including the scaled integer problem in the mixed-integer reduction.

The following standard determinant bounds specialize the polyhedral encoding estimates discussed in [13, Chapter 10]. We include the short argument to record the explicit common exponent Φ used below.

Lemma 2.1 (Coordinates and slacks)**.** *If $P \ne \varnothing$, every $\mathbf{x} \in P$ satisfies $|x_j| < 2^\Phi$ for every j. Every shifted slack satisfies $1/2 \le s_i(\mathbf{x}) < 2^\Phi$, with Φ defined in* (6)*.*

Proof. A vertex of a bounded polyhedron in $\mathbb{R}^d$ has d linearly independent active constraint normals, including when the polyhedron is lower-dimensional. Cramer's rule expresses each vertex coordinate as a ratio of two integral determinants of order d. The nonzero denominator has absolute value at least one. Since every entry of $A, \mathbf{b}$ has absolute value less than 2^τ, the determinant expansion gives

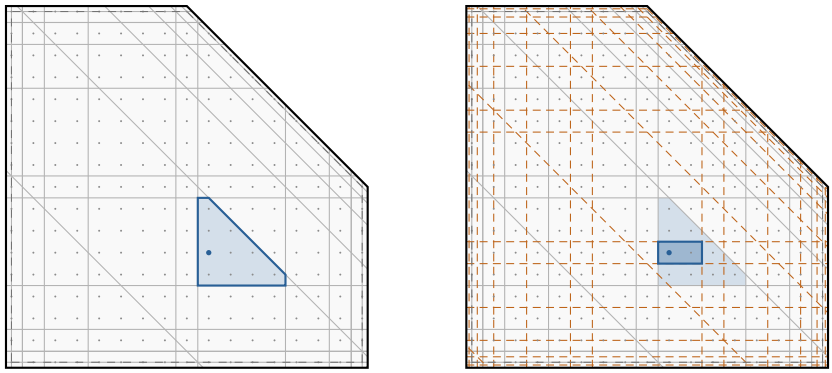
$$|x_j| \le d! 2^{d\tau} \le 2^{d(\tau + \lceil \log_2(d+1) \rceil)} < 2^\Phi$$

for every vertex. The same bound holds throughout P, since every point is a convex combination of vertices. Feasibility gives shifted slacks at least $1/2$; substituting the sharper coordinate estimate above gives

$$\begin{aligned} s_i(\mathbf{x}) &\le \tfrac{1}{2} + |b_i| + \sum_{j=1}^{d} |a_{ij}|\,|x_j| \\ &\le \tfrac{1}{2} + 2^{(d+1)(\tau + \lceil \log_2(d+1) \rceil) + 1} < 2^\Phi. \end{aligned}$$

□

The shifted slacks remove the zero-slack boundary case, even when P is lower-dimensional. The dyadic construction in [5, Theorem 2.2], tracing back to [4], groups comparable positive slacks; we refine these groups to obtain long progressions. Figure 2 illustrates the shifted construction.



(a) Dyadic chambers (b) Refined chambers, $\ell = 2$

Figure 2: Closed shifted-slack chambers for $P = \{\mathbf{x} \in [0,16]^2 : x_1 + x_2 \le 24\}$ (dashed outline). The solid outline is Q, whose lower shifted-slack bound is $1/4$; the upper bound is inactive in this example. The fifth shifted slack is $24.5 - x_1 - x_2$, giving the diagonal lines. The shaded dyadic chamber is refined in (b); its darker closed rectangle $[8.5, 10.5] \times [4.5, 5.5]$ contains the representative $(9, 5)$. All lattice points of P, including its boundary points, lie inside Q. Adjacent closed cells share their boundaries.

Definition 2.1 (Closed refined cells)**.** For integers $\Phi, \ell \ge 1$, let

$$\mathcal{B}_{\Phi,\ell} = \{2^\Phi\} \cup \left\{ 2^k \left(1 + \frac{j}{\ell}\right) : -2 \le k < \Phi,\ 0 \le j < \ell \right\}. \tag{9}$$

There are $(\Phi + 2)\ell + 1$ thresholds, starting at $1/4$. Set

$$Q = \{\mathbf{x} : 1/4 \le s_i(\mathbf{x}) \le 2^\Phi \ (1 \le i \le m)\}.$$

Subdivide Q by the hyperplanes $s_i(\mathbf{x}) = a$, $a \in \mathcal{B}_{\Phi,\ell}$. The family $\mathcal{C}$ consists of the closures of the resulting full-dimensional chambers. If Q has empty interior, set $\mathcal{C} = \varnothing$.

Every integer point of P lies in the interior of Q, by Lemma 2.1, and therefore belongs to a closed cell. The cells may overlap on subdivision hyperplanes, but no separate lower-dimensional faces are needed. Conversely, $Q \subseteq \widetilde{P}$, so every integer point in a cell belongs to P. These statements also justify declaring integer infeasibility when Q has empty interior. The set Q is bounded: its recession directions satisfy $A\mathbf{v} = 0$, which forces $\mathbf{v} = 0$ since P is nonempty and bounded.

If $a < b$ are successive thresholds, then

$$b - a \le a/\ell. \tag{10}$$

Indeed, on $[2^k, 2^{k+1}]$ the spacing is $2^k/\ell$ and $a \ge 2^k$. The same argument applies to $[1/4, 1/2]$ and $[1/2, 1]$.

Lemma 2.2 (Refined displacement cover)**.** *Each $C \in \mathcal{C}$ is described by $a_i \le s_i(\mathbf{x}) \le c_i$ for successive thresholds $a_i < c_i$. Set*

$$D_C = \{\mathbf{v} \in \mathbb{R}^d : |\mathbf{a}_i^\top \mathbf{v}| \le a_i \ (1 \le i \le m)\}.$$

Then $\ell(C - C) \subseteq D_C$ and $C \pm D_C \subseteq \widetilde{P}$. In particular,

$$\mathbf{x} + k(\mathbf{y} - \mathbf{x}) \in P \cap \mathbb{Z}^d \quad (\mathbf{x}, \mathbf{y} \in C \cap \mathbb{Z}^d,\ k \in \mathbb{Z},\ |k| \le \ell). \tag{11}$$

Proof. For $\mathbf{x}, \mathbf{y} \in C$, the weak interval bounds give $|s_i(\mathbf{y}) - s_i(\mathbf{x})| \le c_i - a_i \le a_i/\ell$ and $s_i(\mathbf{x}) \ge a_i$. These prove both inclusions row by row. Since D_C is convex and symmetric, the whole real segment $\mathbf{x} + [-\ell, \ell](\mathbf{y} - \mathbf{x})$ lies in $\widetilde{P}$. If $\mathbf{x}, \mathbf{y}$ and k are integral, its sample is integral and hence belongs to P. Boundedness of D_C follows from $A\mathbf{v} = 0 \Rightarrow \mathbf{v} = 0$. □

Remark 2.3 (Relative slacks characterize extrapolation). Let $\mathbf{x}, \mathbf{y} \in \widetilde{P}$, set $\mathbf{v} = \mathbf{y} - \mathbf{x}$, and let $\ell > 0$. The segments $\mathbf{x} + [-\ell, \ell]\mathbf{v}$ and $\mathbf{y} + [-\ell, \ell]\mathbf{v}$, centered at $\mathbf{x}$ and $\mathbf{y}$ respectively, both lie in $\widetilde{P}$ if and only if, for every row i,

$$\ell\,|s_i(\mathbf{y}) - s_i(\mathbf{x})| \le \min\{s_i(\mathbf{x}), s_i(\mathbf{y})\}.$$

Indeed, the minimum slack along the first segment is $s_i(\mathbf{x}) - \ell|s_i(\mathbf{y}) - s_i(\mathbf{x})|$, and similarly for the second. For positive slacks the condition is equivalently

$$\frac{\max\{s_i(\mathbf{x}), s_i(\mathbf{y})\}}{\min\{s_i(\mathbf{x}), s_i(\mathbf{y})\}} \le 1 + \frac{1}{\ell};$$

if either slack is zero, both must be zero. Our cells have positive shifted slacks, so this last case never arises. Thus multiplicative slack control is exactly the geometry needed for extrapolation about both points. Refined dyadic intervals enforce this condition with rational inequalities of polynomial encoding size, and the integer slack bounds limit the number of dyadic intervals needed. The condition is independent of the objective: Proposition 2.7 turns the resulting cover into a simultaneous approximation guarantee for every polynomial of degree at most D.

The integer progression property is the geometric input to approximation. Closed chambers are defined by non-strict linear inequalities; rounding the slack bounds below removes threshold denominators without changing the integer points.

Lemma 2.4 (Number, encoding, and enumeration of cells)**.** *Let* $H = m((\Phi + 2)\ell + 1)$ *and set*

$$B = \tau + \Phi + \lceil \log_2(\ell + 1) \rceil + 10.$$

There are at most $R_d(H)$ *closed cells. Each has at most* $2m$ *non-strict slack inequalities with coefficient bit length* $O(B)$*. They can be enumerated deterministically in*

$$(H + 1)R_d(H)\ \mathrm{poly}(\langle A, \mathbf{b} \rangle + d + m + B) \tag{12}$$

bit operations. Each cell C *contains a rational polytope* $\widehat{C} \subseteq C$ *with exactly the same integer points, described by at most* $2m$ *inequalities of coefficient bit length* $O(\tau + \Phi + 1)$*.*

Proof. The standard chamber bound for H hyperplanes in $\mathbb{R}^d$ is $\sum_{j=0}^{d} \binom{H}{j}$; see [14]. Indeed, insertion of one hyperplane creates at most as many new chambers as the induced arrangement on that hyperplane, giving the recurrence $R_d(H) = R_d(H-1) + R_{d-1}(H-1)$ for the upper bound. Restriction to the convex set Q cannot split a chamber, so the same bound applies here.

Each threshold has denominator dividing 4ℓ and numerator of bit length $O(\Phi + \log(\ell + 1))$. Clearing this denominator and the half-unit shift gives the stated coefficient bound.

Start with Q and insert the threshold hyperplanes one at a time. Split each current closed cell into its two closed halfspace intersections, keeping only those with nonempty interior. Their closures still cover the parent: if the hyperplane cuts its interior, both pieces have interior; otherwise one piece is the parent. Thus no separate equality branch is needed. At most $R_d(H)$ cells occur at any stage. Bounds from the same row compress to the strongest lower and upper bounds, leaving at most $2m$ inequalities per cell.

For a system $u_j^\top \mathbf{x} \le v_j$ with nonzero normals, test nonempty interior by maximizing δ subject to $u_j^\top \mathbf{x} \le v_j - \delta$ and $0 \le \delta \le 1$. The optimum is positive exactly when the system has interior. Zero rows were removed in preprocessing. These are rational LPs in $d+1$ variables, and there are $O((H+1)R_d(H))$ tests, including the initial test for Q.

Finally, replace $a \le s_i(\mathbf{x}) \le c$ by

$$\lceil a - \tfrac{1}{2} \rceil \le b_i - \mathbf{a}_i^\top \mathbf{x} \le \lfloor c - \tfrac{1}{2} \rfloor.$$

The middle expression is integral on $\mathbb{Z}^d$, so this defines $\widehat{C} \subseteq C$ with the same integer points. Its right-hand sides have $O(\tau + \Phi + 1)$ bits. All rounding is exact. Cells can overlap, so representatives may be repeated; this does not affect feasibility or selection. □

Remark 2.5 (Parallel families). If row i contributes q_i distinct threshold hyperplanes, the count can be sharpened to

$$|\mathcal{C}| \le \sum_{\substack{I \subseteq \{1,\dots,m\} \\ |I| \le d}} \prod_{i \in I} q_i \le \sum_{j=0}^{d} \binom{m}{j} ((\Phi+2)\ell + 1)^j.$$

The insertion argument counts intersections involving at most one hyperplane from each parallel family; further dependencies can only reduce the count. This improves the bound without changing its worst-case ℓ^d dependence. We retain $R_d(H)$ for simplicity. The boundary-vertex count in [4, pp. 30–31] bounds integer-hull vertices, not all reflecting cells, and therefore does not directly replace this count.

We now turn to the objective. A polynomial can vary substantially between consecutive integer samples, so bounded sample values do not immediately bound its derivatives. Markov's inequality resolves this in two steps: first it controls the polynomial between the samples, and then symmetrization reduces the needed second difference to a first difference of a lower-degree polynomial. At an integer minimizer, the backward sample supplies the additional inequality needed to turn that derivative bound into an approximation guarantee. This avoids imposing a continuous stationarity condition or approximating the objective uniformly on every cell.

The following elementary consequence of the classical Markov inequality combines the standard sample-to-supremum argument used in polynomial norming meshes (see [8, proof of Proposition 2]) with an even-part bound for a centered second difference. We include the short proof for the discrete-minimum formulation needed here.

We use the classical Markov inequality [7, Theorem 5.1.8]: for a real polynomial q of degree at most D and an interval $[u, v]$ with $u < v$,

$$\max_{u \le t \le v} |q'(t)| \le \frac{2D^2}{v-u} \max_{u \le t \le v} |q(t)|.$$

Thus the largest absolute derivative is bounded by the largest absolute polynomial value, multiplied by a factor depending only on the degree and interval length.

Bounds between equally spaced samples go back to Ehlich and Zeller [15]; Coppersmith and Rivlin [16] quantify their dependence on degree and sample count, with sharp fixed-ratio asymptotics given by Kuijlaars [17]. The reduction of symmetric-node problems to polynomials in t^2 also appears in Rivlin [18, p. 539], for coefficient bounds. We retain the elementary proof below for the discrete-minimum estimate needed here; these references concern its classical ingredients.

Lemma 2.6 (Approximation at a discrete minimum)**.** *Let $D \geq 1$ and $\ell \geq D^2$ be integers, and let $p \in \mathbb{R}[t]$ have degree at most D. Suppose*

$$a \leq p(k) \leq b \quad (k = -\ell, \dots, \ell), \qquad p(-1) \geq p(0).$$

Then

$$p(1) - p(0) \leq \frac{D^2}{\ell^2}(b - a).$$

Proof. Set $q = p - (a+b)/2$ and $W = \max_{-\ell \leq t \leq \ell} |q(t)|$. Markov's inequality gives $|q'(t)| \leq D^2 W/\ell$ for every $t \in [-\ell, \ell]$. Every point of $[-\ell, \ell]$ lies within $1/2$ of a sampled integer, so

$$W \leq \frac{b-a}{2} + \frac{D^2}{2\ell} W \leq \frac{b-a+W}{2}, \qquad W \leq b - a.$$

Let $n = \lfloor D/2 \rfloor$. The even part of q has the form

$$e(t) = \frac{q(t) + q(-t)}{2} = h(t^2), \qquad \deg h \leq n.$$

Thus $|h(t)| \leq W$ on $[0, \ell^2]$. Markov's inequality gives $|h'(t)| \leq 2n^2 W/\ell^2$ throughout that interval. Since $\ell \geq 1$, the mean value bound on $[0, 1]$ and $p(-1) \geq p(0)$ yield

$$\begin{aligned} p(1) - p(0) &\leq p(1) + p(-1) - 2p(0) \\ &= 2(h(1) - h(0)) \\ &\leq \frac{4\lfloor D/2 \rfloor^2}{\ell^2}(b-a) \leq \frac{D^2}{\ell^2}(b-a). \end{aligned}$$

For $D = 1$, h is constant and the bound is zero. The argument also covers $a = b$. □

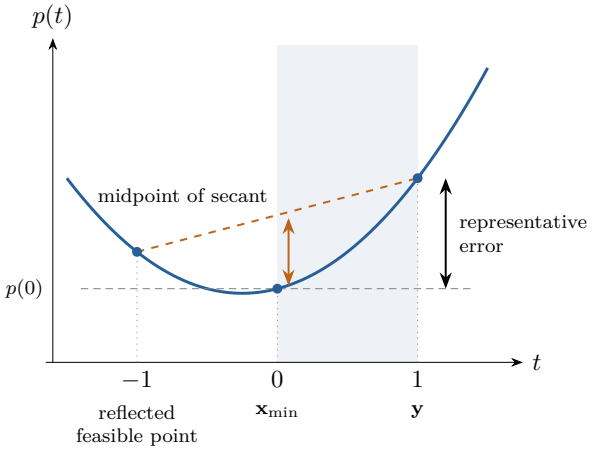
The estimate is needed only on the cell containing a global minimizer, and holds for any representative of that cell. Applying it also to $-f$ gives the following simultaneous approximation of both extrema. The general conversion from a polynomial norming inequality to range-relative optimization is given in [8, Proposition 1]; here we use the discrete-minimum estimate directly. The reflecting geometry supplies the feasible integer samples needed here.

Proposition 2.7 (Representatives approximate extrema)**.** *Choose one integer representative from each integer-feasible closed cell, forming Y. If $P \cap \mathbb{Z}^d \neq \varnothing$ and $\ell \geq D^2$, then every real polynomial f of degree at most D satisfies*

$$0 \leq \min_{Y} f - \min_{P \cap \mathbb{Z}^d} f \leq \frac{D^2}{\ell^2}(f_{\max} - f_{\min}),$$
$$0 \leq \max_{P \cap \mathbb{Z}^d} f - \max_{Y} f \leq \frac{D^2}{\ell^2}(f_{\max} - f_{\min}).$$

Proof. Let $\mathbf{x}_{\min}$ minimize f on $P \cap \mathbb{Z}^d$, and let $\mathbf{y}$ represent a cell containing it. Put $p(t) = f(\mathbf{x}_{\min} + t(\mathbf{y} - \mathbf{x}_{\min}))$. Lemma 2.2 makes all integer samples with $|k| \leq \ell$ feasible, so they lie between $f_{\min}$ and $f_{\max}$. Minimality gives $p(-1) \geq p(0)$. Lemma 2.6 now bounds $f(\mathbf{y}) - f(\mathbf{x}_{\min})$. Apply the same argument to $-f$ for the maximum. □

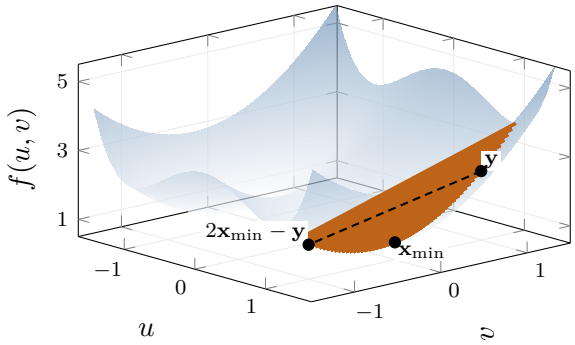
(a) One-dimensional comparison



$$p(t) = f(\mathbf{x}_{\min} + t(\mathbf{y} - \mathbf{x}_{\min})),$$

$$\begin{aligned} p(-1) &\geq p(0), \\ p(1) - p(0) &\leq p(1) + p(-1) - 2p(0) \\ &\leq \|p''\|_{\infty,[-\ell,\ell]}. \end{aligned}$$

(b) Nonconvex quartic and its slice



$$f(u,v) = 1 + (u^2 - 1)^2 + v^2 + \tfrac{1}{2}v,$$
$$f(1,t) = 1 + t^2 + \tfrac{1}{2}t.$$

Figure 3: Polynomial approximation along a feasible direction. (a) The shaded segment joins an integer minimizer to its cell representative. The reflected point is feasible, and the secant midpoint gap is half the symmetric second difference. (b) A local view of a nonconvex quartic: $f_{uu}(0,v) = -4$. On $[-16,16]^2 \cap \mathbb{Z}^2$, $\mathbf{x}_{\min} = (1,0)$ is a global minimizer. The orange slice through $\mathbf{y} = (1,1)$ is exactly the polynomial in (a); black dots mark the three integer samples. The surface illustrates the analytic comparison, not a particular dyadic cell. The full progression for $|t| \leq 16 = D^2$ extends beyond the displayed patch. Neither the proof nor the illustration assumes continuous stationarity at $\mathbf{x}_{\min}$.

3 Algorithm and complexity

The preceding results lead directly to Algorithm 1: enumerate the refined cells, find one integer point in each nonempty cell, and compare the objective values at those points. The geometric work is independent of the objective coefficients. This is also why the representatives can be reused for different polynomials of degree at most D. For this purpose, the algorithm accepts any prescribed degree bound $D \geq \max\{1, \deg f\}$; Theorem 1.1 uses the bound computed from f. A constant objective requires only one integer feasibility query; dimension zero is similarly immediate. Representatives can be processed as a stream, or saved as a universal set Y for subsequent objectives.

Refined-cell sampling SampleMin(P, f, D, ε)

Input: bounded rational $P \subseteq \mathbb{R}^d$, $f \in \mathbb{Z}[\mathbf{x}]$, $\deg f \leq D$, $D \geq 1$, and $0 < \varepsilon \leq 1$.
Output: $\mathbf{x}_\varepsilon \in P \cap \mathbb{Z}^d$, or INFEASIBLE.

1 Clear row denominators: $P = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$, $A, \mathbf{b}$ integral.
2 **if** $P = \varnothing$ **then return** INFEASIBLE; otherwise remove zero constraint rows.
3 Compute Φ from (6); set $\ell \leftarrow \max\{D^2, \lceil D/\sqrt{\varepsilon} \rceil\}$, $Y \leftarrow \varnothing$.
4 Construct the thresholds $\mathcal{B}_{\Phi,\ell} \leftarrow \{2^\Phi\} \cup \{2^k(1 + j/\ell) : \ -2 \leq k < \Phi,\ 0 \leq j < \ell\}$.
5 Enumerate the closed chambers $\mathcal{C}$ of $s_i(\mathbf{x}) = a$, $a \in \mathcal{B}_{\Phi,\ell}$, as in Definition 2.1.
6 **for each** $C \in \mathcal{C}$ **do**
7 Form $\widehat{C} \subseteq C$ with $\widehat{C} \cap \mathbb{Z}^d = C \cap \mathbb{Z}^d$ by rounding shifted slack bounds (Lemma 2.4).
8 **if** integer linear feasibility returns $\mathbf{z}_C \in \widehat{C} \cap \mathbb{Z}^d$ **then** $Y \leftarrow Y \cup \{\mathbf{z}_C\}$.
9 **if** $Y = \varnothing$ **then return** INFEASIBLE.
10 **return** $\mathbf{x}_\varepsilon \in \arg\min_{\mathbf{y} \in Y} f(\mathbf{y})$, using exact evaluation.

Guarantee: $0 \leq f(\mathbf{x}_\varepsilon) - f_{\min} \leq \varepsilon(f_{\max} - f_{\min})$.

Algorithm 1: The representative algorithm. Steps 1–9 depend only on P, D, ε; the objective coefficients enter in step 10. Replacing arg min by arg max gives the corresponding maximization guarantee.

Correctness in Theorem 1.1. Every integer-feasible cell contributes a feasible representative, and the closed cells cover $P \cap \mathbb{Z}^d$. Thus $Y = \varnothing$ if and only if $P \cap \mathbb{Z}^d = \varnothing$. Otherwise, Proposition 2.7 and $\ell = \max\{D^2, \lceil D/\sqrt{\varepsilon} \rceil\}$ give

$$0 \leq \min_Y f - f_{\min} \leq \frac{D^2}{\ell^2}(f_{\max} - f_{\min}) \leq \varepsilon(f_{\max} - f_{\min}).$$

The algorithm returns a minimizer on Y. The cover and representatives depend only on P, D, ε, so the same set works for every polynomial of degree at most D. □

It remains to account for bit operations. There are three costs: constructing the arrangement, producing integer representatives, and evaluating the objective. The distinction between the number of cells and the sizes of their descriptions is important: refinement increases the former polynomially in $1/\sqrt{\varepsilon}$, whereas it increases the threshold bit lengths only logarithmically. By Lemma 2.4, chamber enumeration costs (12). The bit length of ℓ is $O(\log(D+1)+\langle\varepsilon\rangle)$; compute the least positive integer k satisfying $\varepsilon k^2 \geq D^2$ by binary search with rational comparisons, and set $\ell = \max\{D^2, k\}$. Generating all thresholds uses $O(\Phi\ell)$ arithmetic operations on integers with $O(\Phi + \log(\ell+1))$ bits and is included in the arrangement bound when $m \geq 1$. A nonempty bounded polyhedron in positive dimension cannot be specified by zero inequalities.

Once a cell has been enumerated, its integer feasibility query has a description whose coefficient sizes do not grow with the refinement parameter. Each of at most $R_d(H)$ queries has d integer variables and a total encoding size

$$S_C = O(m(d+1)(\tau+\Phi+1)+d+m)\,. \tag{13}$$

Fixed-dimensional integer linear feasibility with witness production takes $2^{O(d\log(d+1))}\,\mathrm{poly}(S_C)$ bit operations using Kannan's algorithm [19], improving the dimension dependence of Lenstra's algorithm [20]. Every integer point returned from a shifted cell belongs to the original bounded polytope. Accordingly, every returned coordinate has at most $\Phi + O(1)$ bits by Lemma 2.1; its bit length grows logarithmically with its magnitude. Notice also that (13) has no dependence on ℓ other than the number of cells, because the threshold inequalities have already been rounded to integer slack bounds.

Finally, the coordinate bound ensures that exact objective evaluation also has polynomial bit cost, even though feasible coordinates can be numerically large. The integer coefficients of f have bit length at most $\langle f\rangle$. If $\mathbf{x} \in P \cap \mathbb{Z}^d$, every monomial has absolute value at most $2^{D\Phi}$. Each value of $f(\mathbf{x})$, and the intermediates obtained by evaluating monomials and summing them, has $O(\langle f\rangle + D\Phi + \log(|\mathcal{A}|+1))$ bits. Repeated squaring evaluates the explicit exponent vectors in a number of arithmetic operations polynomial in $\langle f\rangle + d + \log(D+1)$. Even schoolbook integer arithmetic therefore gives

$$\mathrm{poly}(\langle f\rangle + d + D\Phi) \tag{14}$$

bit operations per representative. The case of the zero polynomial is immediate. Returning an exact integer objective value also fits this bound. No unit-cost assumption on rational arithmetic is used.

Combining these three costs gives the explicit bound

$$\begin{aligned} T \leq\ & O((H+1)R_d(H))\ \mathrm{poly}(\varphi, \log(\ell+1)) \\ & + R_d(H)\, 2^{O(d\log(d+1))}\,\mathrm{poly}(\varphi) \\ & + R_d(H)\ \mathrm{poly}(\langle f\rangle + d + D\Phi). \end{aligned} \tag{15}$$

The polynomial here has an absolute exponent. Since $\Phi = O(d(\tau+\log(d+1)))$ and $\Lambda \leq \ell \leq 2\Lambda$ for $\Lambda = \max\{D^2, D/\sqrt{\varepsilon}\}$, the more detailed parameter bound is

$$2^{O(d\log(d+1))}\,(1 + md(\tau+\log(d+1))\Lambda)^{d+1}\,\mathrm{poly}(\varphi + Dd(\tau+\log(d+1)))\,.$$

For the integral input in Theorem 1.1, $\tau = O(\varphi)$ and $d \leq \varphi$. Consequently $\tau + \log(d+1) = O(\varphi)$, and the factor d^{d+1} is absorbed into $2^{O(d\log(d+1))}$. The polynomial overhead is bounded by $\mathrm{poly}(\varphi, D)$. Since $m, \varphi, D \geq 1$ and $\varepsilon \leq 1$, the additive one can also be absorbed, giving

$$2^{O(d\log(d+1))}\,(m\varphi\Lambda)^{d+1}\,\mathrm{poly}(\varphi, D).$$

Finally, $m, d \le \varphi$ allows the dimension factor and the polynomial overhead to be absorbed into $(\varphi D/\varepsilon)^{O(d)}$, proving (3). The dependence on the objective encoding $\langle f \rangle$ enters through exact evaluation in (14); the cell construction is independent of the objective coefficients. The factor $2^{O(d\log(d+1))}$ comes from integer linear feasibility. The exponent $d+1$ includes the extra factor from incremental chamber enumeration. For fixed d, normalization and all encoding terms are polynomial in the original φ and numerical D; the accuracy encoding also bounds $\log(\ell+1)$ up to a term $O(\log(D+1))$. This proves Theorem 1.1. The algorithm makes at most $R_d(H)$ integer-feasibility queries and $O((H+1)R_d(H))$ LP enumeration tests. For fixed P, D, d, these counts are $O(\varepsilon^{-d/2})$ and $O(\varepsilon^{-(d+1)/2})$, respectively; they are not bit-operation bounds.

Proof of Corollary 1.2. Applying Proposition 2.7 to $-f$ gives $f_{\max} - f(\mathbf{x}_\varepsilon) \le \varepsilon(f_{\max} - f_{\min})$. Nonnegativity on $P \cap \mathbb{Z}^d$ implies $f_{\max} - f_{\min} \le f_{\max}$. □

Proof of Corollary 1.3. If $d_c = 0$, apply Theorem 1.1. Otherwise let $S = P \cap (\mathbb{R}^{d_c} \times \mathbb{Z}^{d_i})$. By [1, Lemma 9], an integer $\Delta > 0$ of polynomial encoding size in fixed dimension can be computed such that every nonempty slice $\Delta P_{\mathbf{z}}$ is integral. Our modification begins with their barycentric rounding [1, Lemma 10]: instead of rounding down all but one weight, we round up all but a largest weight. This gives the controlled feasible segment below, on which Markov's inequality replaces their coefficient-based Lipschitz estimate [1, Lemma 11]. Set

$$k = \left\lceil \frac{4D^2 d_c(d_c+1)}{\varepsilon} \right\rceil, \qquad Q = k\Delta, \qquad t = \frac{d_c(d_c+1)}{k}.$$

For any $\mathbf{x} \in P_{\mathbf{z}}$, Carathéodory's theorem gives $\mathbf{x} = \sum_{i=0}^{d_c} \lambda_i \mathbf{v}_i$ for slice vertices $\mathbf{v}_i \in \Delta^{-1}\mathbb{Z}^{d_c}$ and convex weights, indexed so that $\lambda_0 \ge 1/(d_c+1)$. Put

$$\mu_i = \frac{\lceil k\lambda_i \rceil}{k} \ (i \ge 1), \qquad \mu_0 = 1 - \sum_{i=1}^{d_c} \mu_i.$$

For $i \ge 1$, $\lambda_i \le \mu_i \le \lambda_i + 1/k$, while $\mu_0 \ge \lambda_0 - d_c/k \ge (1-t)\lambda_0$. Since $0 < t \le 1$, all weights are nonnegative and $\mu_i \ge (1-t)\lambda_i$. Therefore

$$\mathbf{y} = \sum_{i=0}^{d_c} \mu_i \mathbf{v}_i = (1-t)\mathbf{x} + t\mathbf{w} \in P_{\mathbf{z}} \cap Q^{-1}\mathbb{Z}^{d_c}$$

for some $\mathbf{w} \in P_{\mathbf{z}}$: its convex weights are $(\mu_i - (1-t)\lambda_i)/t$.

Write $R = f_{\max} - f_{\min}$ over S. The polynomial $p(u) = f((1-u)\mathbf{x} + u\mathbf{w}, \mathbf{z}) - f_{\min}$ has degree at most D and lies in $[0, R]$ on $[0, 1]$. Markov's inequality gives

$$|f(\mathbf{y}, \mathbf{z}) - f(\mathbf{x}, \mathbf{z})| \le 2D^2 tR \le \varepsilon R/2.$$

Thus the grid preserves nonemptiness and approximates the minimum within $\varepsilon R/2$, including when $R = 0$. Apply Theorem 1.1 on the grid with accuracy $\varepsilon/2$. Its objective range is at most R, so the total error is at most εR.

Substitute $\mathbf{x} = \mathbf{u}/Q$ and use the integer polynomial $g(\mathbf{u}, \mathbf{z}) = Q^D f(\mathbf{u}/Q, \mathbf{z})$. Its degree and support are unchanged, and coefficient bit lengths increase by $O(D\log(Q+1))$. Since

$$\log Q = \log \Delta + O(\log(D+1) + \log(d_c+1) + \log(1/\varepsilon)),$$

positive scaling preserves the guarantee; negation and nonnegativity give the maximization claim. For explicit complexity, normalize the inequalities to integers and let τ bound their coefficient bit

lengths. DHKW construct Δ as the least common multiple of nonzero absolute determinants of d_c-row bases of the continuous-variable matrix [1, proof of Lemma 9]. Hadamard's inequality gives

$$\log \Delta = O\left(\binom{m}{d_c} d_c(\tau + \log(d_c + 1))\right).$$

Enumerating these bases and computing their least common multiple takes $\varphi^{O(d_c+1)}$ bit operations; normalization changes encoding size only polynomially. The scaled input has size at most $\varphi^{O(d_c+1)}$ under our unary-exponent convention. Applying Theorem 1.1 proves (4). For binary exponents, the extra polynomial dependence on D is likewise absorbed into the displayed bound.

The new grid multiplier avoids the coefficient Lipschitz constant and positive range-gap bound in DHKW's reduction. This sharpens the sufficient grid resolution, but does not alone establish a better overall asymptotic runtime: the integer solvers depend on $\log Q$, and the common denominator Δ remains. □

4 Discussion and extensions

Symmetrization gives two refinement regimes: $\ell = D^2$ when $\varepsilon \geq D^{-2}$, and $\ell = \lceil D/\sqrt{\varepsilon} \rceil$ when $\varepsilon \leq D^{-2}$. In the latter regime, replacing the earlier twice-differentiated Markov estimate improves the refinement-dependent factors in the cell-count and enumeration bounds by D^d and D^{d+1}, respectively, up to ceiling effects. These are improvements in upper bounds, not guaranteed instance-by-instance speedups; the accuracy exponents are unchanged.

The accuracy exponent for objective-independent representative sets has a simple cardinality obstruction. On $P_N = [0, N]^d$, suppose $Y \subseteq P_N \cap \mathbb{Z}^d$ approximates all quadratic minima with range-relative accuracy ε. For each integer center $\mathbf{a}$, apply this guarantee to $f_{\mathbf{a}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{a}\|_2^2$. Its minimum is zero and its range is at most dN^2, so some $\mathbf{y} \in Y$ lies within $N\sqrt{d\varepsilon}$ of $\mathbf{a}$. Each representative covers at most $(2N\sqrt{d\varepsilon} + 1)^d$ integer centers, giving

$$|Y| \geq \frac{(N+1)^d}{(2N\sqrt{d\varepsilon} + 1)^d}.$$

For integer $N \geq \varepsilon^{-1/2}$ this is $\Omega_d(\varepsilon^{-d/2})$. This lower bound concerns universal representative sets across a family of boxes, not arbitrary optimization algorithms or one fixed finite lattice set. It does not explain the additional $\varepsilon^{-1/2}$ factor in our enumeration bound.

For any refinement $\ell \geq 1$, the representatives include every vertex of the integer hull. Indeed, if such a vertex $\mathbf{v}$ shared a cell with a distinct integer point $\mathbf{y}$, reflection would make $2\mathbf{v} - \mathbf{y}$ feasible, contradicting extremality of $\mathbf{v} = (\mathbf{y} + (2\mathbf{v} - \mathbf{y}))/2$. Thus every linear objective is optimized exactly.

The main result concerns polynomial optimization: a refined reflecting cover and one integer-feasibility query per cell suffice for a range-relative FPTAS. Corollary 1.3 supplies the mixed-integer extension through the established grid reduction. Table 1 places this construction alongside related decomposition and sampling methods; the common error-transfer principle is recorded in Appendix A.

For polynomial objectives with additional structure, one can replace feasibility by optimization of a local surrogate h_C. If $|f - h_C| \leq \delta_C$ on the feasible part of a cell and a feasible candidate minimizes h_C there within additive error η_C, then selecting the best candidate gives error at most $2\delta_{C^*} + \eta_{C^*}$, where C^* contains a global minimizer. Affine, convex, or quadratic surrogates are useful when their local optimization problems are tractable and their errors can be bounded relative to the global objective range. This includes the frozen-weight idea for sliceable objectives [10] and affine

Method	Comparison regions	Local procedure	Error certificate
Reflecting cells (this paper)	Refined geometric slack cells	Integer feasibility, $h_C = 0$	Reflection and sampled Markov bound at an optimum
Geometric product sampling	Multiplicative coordinate boxes	Feasible representative	Bounded variation of $\log f$
Sliceable functions [10]	Geometric affine slices	Optimize frozen-weight objective	Multiplicative sandwich, with sign conditions
Approximate Pareto methods [9, 21, 22]	Geometric threshold regions	Gap or dominance oracle	Coordinatewise dominance and aggregate scaling
Product of two linear costs [23]	Geometric budget regions	Parametric LP and extreme-point recovery	Budget error and product comparison
Quadratic mesh methods [3, 24]	Boxes after problem-specific preprocessing	MILP or ILP for affine model	Uniform residual bound relative to objective range
Simplex grids and norming meshes [25, 26, 8]	Neighborhoods assigned to samples	Evaluate designated sample	Grid or norming estimate; second-order bounds where available

Table 1: Related decomposition and sampling approaches. Each row retains the hypotheses and fixed parameters of the cited work. These connections do not assert identical partitions or approximation guarantees.

models for quadratics [3]. An FPTAS still requires polynomial bounds on the cover construction, local computations, and encoding sizes; the transfer inequality alone does not supply them.

A further direction is to exploit a supplied factorization $f(\mathbf{x}) = g(B\mathbf{x}) + \mathbf{c}^\top \mathbf{x}$, where $B \in \mathbb{Q}^{k\times d}$ has few rows. Local affine optimization could handle the remaining directions without refinement, suggesting an accuracy dependence governed by k rather than d, as in the quadratic setting of Del Pia [3]. For higher-degree polynomials, this would require a cover whose size reflects the nonlinear dimension while preserving sufficient feasible samples to control the approximation error relative to the integer objective range. Even with fixed ambient dimension, replacing the current accuracy dependence by $\varepsilon^{-O(k)}$ is not established by our analysis; merely changing the local oracle leaves the existing cell count unchanged.

The reflecting argument also extends to polynomial optimization over a bounded set K whenever an efficiently constructible polynomial-size cover of $K \cap \mathbb{Z}^d$ supports the same feasible integer progressions and admits efficient integer-feasibility queries. An outer polyhedral relaxation alone is insufficient, because reflected points may leave K; a suitable relaxation with exactly the same integer points would suffice. General convex sets need not admit such a cover: at a boundary lattice point of a strictly convex set, the reflection property forces its cell to contain no other lattice point. Extensions beyond polyhedra may therefore require different local error estimates or stronger local optimization oracles. These are conditions for further generalization, rather than additional FPTAS results established here.

For continuous extensions, Pierzchała [27, Theorem 1.4] proves that polynomial images of compact Markov sets remain Markov sets when the map has full generic rank in the target. Combined with [8, Proposition 2], this supplies norming meshes on further continuous sets. It does not supply integer representatives or uniform encoding bounds for the Markov constants, and hence does not by itself extend our integer FPTAS.

A A local comparison principle

Let $\mathcal{X}$ denote an abstract feasible set and let F be the objective to minimize. For maximization use $F = -f$; for strictly positive objectives, $F = \log f$ or $F = -\log f$ expresses multiplicative guarantees additively. These transformations are analytical: candidates may be compared using the original objective, without computing logarithms. A cover procedure supplies finitely many comparison regions C, admissible output sets $R_C \subseteq \mathcal{X}$, and surrogate functions h_C defined on $(C \cap \mathcal{X}) \cup R_C$. Usually $R_C = C \cap \mathcal{X}$, but allowing a larger output set accommodates dominance oracles and rounding.

Construct the finite cover and, for each region C, either certify $C \cap \mathcal{X} = \varnothing$ or return a feasible candidate $y_C \in R_C$ with

$$h_C(y_C) \leq \inf_{x \in C \cap \mathcal{X}} h_C(x) + \eta_C.$$

Report infeasibility if no candidate is returned; otherwise select a candidate with minimum F-value.

Proposition A.1 (Local comparison principle)**.** *Suppose $\mathcal{X} \neq \varnothing$, F attains its minimum at x^*, and the cover and local oracles in the procedure above satisfy their stated guarantees, with $\eta_C \geq 0$. Suppose some $C^* \ni x^*$ satisfies, for every $y \in R_{C^*}$,*

$$(F - h_{C^*})(y) - (F - h_{C^*})(x^*) \leq \beta_{C^*}.$$

Then the returned point satisfies

$$F(\widehat{x}) - F(x^*) \leq \eta_{C^*} + \beta_{C^*}.$$

If $\mathcal{X} = \varnothing$, the algorithm reports infeasibility.

Proof. The oracle cannot discard C^*, since it contains x^*. By selection of $\widehat{x}$,

$$F(\widehat{x}) - F(x^*) \leq F(y_{C^*}) - F(x^*) \leq h_{C^*}(y_{C^*}) - h_{C^*}(x^*) + \beta_{C^*} \leq \eta_{C^*} + \beta_{C^*}.$$

When $\mathcal{X} = \varnothing$, no feasible candidate can be returned and every region is discarded. □

The algorithm has cost

$$T_{\text{cover}} + \sum_{C \in \mathcal{C}} (T_{\text{local}}(C) + T_{\text{compare}}(C)),$$

where comparison includes objective evaluation. An FPTAS follows only when these costs and all constructed descriptions are polynomial in the input size and $1/\varepsilon$, for the fixed structural parameters, and the error bound gives the requested guarantee. The scale $\varepsilon(f_{\max} - f_{\min})$ or a logarithmic multiplicative error may be used in the proof without being known numerically to the algorithm. Establishing effective covers, local oracles, and error certificates remains the substantive problem in each application.

References

[1] J. A. De Loera, R. Hemmecke, M. Köppe, and R. Weismantel. FPTAS for optimizing polynomials over the mixed-integer points of polytopes in fixed dimension. *Mathematical Programming* 115 (2008), 273–290. `https://doi.org/10.1007/s10107-007-0175-8`.

[2] J. A. De Loera, R. Hemmecke, M. Köppe, and R. Weismantel. Integer polynomial optimization in fixed dimension. *Mathematics of Operations Research* 31 (2006), 147–153. https://doi.org/10.1287/moor.1050.0169.

[3] A. Del Pia. An approximation algorithm for indefinite mixed integer quadratic programming. *Mathematical Programming* 201 (2023), 263–293. https://doi.org/10.1007/s10107-022-01907-3.

[4] W. Cook, M. Hartmann, R. Kannan, and C. McDiarmid. On integer points in polyhedra. *Combinatorica* 12 (1992), 27–37. https://doi.org/10.1007/BF01191202.

[5] C. Ari and R. Hildebrand. *Hidden Convexity via Symmetric Displacement Covers: Mixed-Integer Quadratic Programming and Integer Cubic Programming in Fixed Dimension.* arXiv:2609.18266, version 2, September 23, 2026. https://arxiv.org/abs/2609.18266v2.

[6] L. Wei. *Integer quadratic programming in fixed dimension is polynomial-time solvable.* Optimization Online, 2026. https://optimization-online.org/?p=36744.

[7] P. Borwein and T. Erdélyi. *Polynomials and Polynomial Inequalities.* Graduate Texts in Mathematics 161, Springer, New York, 1995. https://doi.org/10.1007/978-1-4612-0793-1.

[8] F. Piazzon and M. Vianello. Markov inequalities, Dubiner distance, norming meshes and polynomial optimization on convex bodies. *Optimization Letters* 13 (2019), 1325–1343. https://doi.org/10.1007/s11590-018-1377-0.

[9] C. H. Papadimitriou and M. Yannakakis. On the approximability of trade-offs and optimal access of Web sources. *Proceedings of the 41st Annual Symposium on Foundations of Computer Science*, 2000, 86–92. https://doi.org/10.1109/SFCS.2000.892068.

[10] R. Hildebrand, R. Weismantel, and K. Zemmer. An FPTAS for minimizing indefinite quadratic forms over integers in polyhedra. *Proceedings of the 2016 Annual ACM–SIAM Symposium on Discrete Algorithms*, 2016, pp. 1715–1723. https://doi.org/10.1137/1.9781611974331.ch118.

[11] A. I. Barvinok. A polynomial time algorithm for counting integral points in polyhedra when the dimension is fixed. *Mathematics of Operations Research* 19 (1994), 769–779. https://doi.org/10.1287/moor.19.4.769.

[12] I. Pak and G. Panova. On the complexity of computing Kronecker coefficients. *Computational Complexity* 26 (2017), 1–36. https://doi.org/10.1007/s00037-015-0109-4.

[13] A. Schrijver. *Theory of Linear and Integer Programming.* Wiley, Chichester, 1986.

[14] R. P. Stanley. An introduction to hyperplane arrangements. In *Geometric Combinatorics*, IAS/Park City Mathematics Series, vol. 13, American Mathematical Society, 2007, pp. 389–496. https://doi.org/10.1090/pcms/013/08.

[15] H. Ehlich and K. Zeller. Schwankung von Polynomen zwischen Gitterpunkten. *Mathematische Zeitschrift* 86 (1964), 41–44. https://doi.org/10.1007/BF01111276.

[16] D. Coppersmith and T. J. Rivlin. The growth of polynomials bounded at equally spaced points. *SIAM Journal on Mathematical Analysis* 23 (1992), 970–983. https://doi.org/10.1137/0523054.

[17] A. B. J. Kuijlaars. Extremal polynomials on the n-grid. *Journal of Approximation Theory* 288 (2023), 105875. https://doi.org/10.1016/j.jat.2023.105875.

[18] T. J. Rivlin. Some properties of polynomials bounded at equally spaced points. In *Approximation and Computation: A Festschrift in Honor of Walter Gautschi*, International Series of Numerical Mathematics 119, Birkhäuser, 1994, pp. 535–542. https://doi.org/10.1007/978-1-4684-7415-2_35.

[19] R. Kannan. Minkowski's convex body theorem and integer programming. *Mathematics of Operations Research* 12 (1987), 415–440. https://doi.org/10.1287/moor.12.3.415.

[20] H. W. Lenstra, Jr. Integer programming with a fixed number of variables. *Mathematics of Operations Research* 8 (1983), 538–548. https://doi.org/10.1287/moor.8.4.538.

[21] S. Mittal and A. S. Schulz. An FPTAS for optimizing a class of low-rank functions over a polytope. *Mathematical Programming* 141 (2013), 103–120. https://doi.org/10.1007/s10107-011-0511-x.

[22] S. Mittal and A. S. Schulz. A general framework for designing approximation schemes for combinatorial optimization problems with many objectives combined into one. *Operations Research* 61 (2013), 386–397. https://doi.org/10.1287/opre.1120.1093.

[23] V. Goyal, L. Genç-Kaya, and R. Ravi. An FPTAS for minimizing the product of two non-negative linear cost functions. *Mathematical Programming* 126 (2011), 401–405. https://doi.org/10.1007/s10107-009-0287-4.

[24] A. Del Pia. Subdeterminants and concave integer quadratic programming. *SIAM Journal on Optimization* 29 (2019), 3154–3173. https://doi.org/10.1137/18M121873X.

[25] E. de Klerk, M. Laurent, and P. A. Parrilo. A PTAS for the minimization of polynomials of fixed degree over the simplex. *Theoretical Computer Science* 361 (2006), 210–225. https://doi.org/10.1016/j.tcs.2006.05.011.

[26] E. de Klerk, M. Laurent, Z. Sun, and J. C. Vera. On the convergence rate of grid search for polynomial optimization over the simplex. *Optimization Letters* 11 (2017), 597–608. https://arxiv.org/abs/1510.01163.

[27] R. Pierzchała. Markov's inequality and polynomial mappings. *Mathematische Annalen* 366 (2016), 57–82. https://doi.org/10.1007/s00208-015-1294-9.

Disclosure of AI assistance and development of the manuscript

Robert Hildebrand conceived and directed this project, using OpenAI's Codex assistant throughout an extended process of mathematical exploration, scrutiny, and revision. AI assistance included mathematical exploration, proposed arguments, literature searches, complexity analysis, and substantial drafting and revision of the LaTeX source. Its role therefore went beyond language editing. The account below records the contributions visible in that exchange; it does not assign priority for the underlying published results or describe the development of the earlier Ari–Hildebrand manuscript.

Hildebrand's contributions. Hildebrand posed the research question: whether the polynomial approximation results of De Loera, Hemmecke, Köppe, and Weismantel could admit a simpler geometric proof, and whether ideas from quadratic approximation could extend to higher degrees. He proposed investigating M-ellipsoid covers and, specifically, the Cook–Hartmann–Kannan–McDiarmid cell decomposition used in the Ari–Hildebrand draft. This suggestion supplied the geometric starting point of the present argument. He directed the assistant to build on that draft, requested complexity bounds in terms of the actual encoding of the problem, and asked for a careful comparison with the earlier approximation schemes, particularly Del Pia's quadratic result.

Hildebrand requested simpler proofs and directed the comparison with the DHKW reduction. He proposed connections to approximate Pareto methods, his earlier quadratic FPTAS, and low-rank optimization, while keeping the final paper centered on polynomial integer optimization.

He supplied historical and contemporary source literature and pressed for direct use of established results, including Schrijver's encoding bounds and the relevant parts of CHKM. He scrutinized the slack shifts, cell boundaries, encoding parameters, and dependence on accuracy, and requested that the resulting complexity statements be explicit and consistent. He initiated the Lean development, directed its expansion from elementary calculations to the geometric and mixed-integer arguments, and requested a clear separation of checked conclusions from published hypotheses and implementation assumptions. He also brought reviews from separate AI-assisted discussions into the development and directed their reconciliation. His role therefore encompassed the mathematical conception, sustained critical examination of the arguments, verification strategy, and selection and presentation of the final results.

AI-assisted development under Hildebrand's direction. In response to this direction, the assistant developed the proposed refined-slack construction and its connection to long feasible integer progressions, drafted the application of the classical Markov inequality to integer samples, and worked out the cell-count and encoding estimates. Earlier drafts used a first-difference bound and then affine Taylor models to improve accuracy dependence. During the simplification discussion, the assistant proposed using the symmetric second difference at a global integer minimizer, exploiting $p(-1) \geq p(0)$. This yielded the present square-root accuracy scaling with arbitrary cell representatives and removed the need for affine-model optimization in the main argument. It also corrected the assistant's earlier assertion that the coarser cover would lose the universal representative guarantee. The assistant searched for related work on polynomial norming meshes and drafted the comparisons and references. A subsequent review supplied by Hildebrand proposed the even-part symmetrization and the one-grid mixed-integer proof; the assistant incorporated these arguments and formalized the strengthened sampled estimate and refinement in Lean. Attribution of these steps to the exchange is not a claim that the mathematical ingredients are new.

Status of Lean verification. The updated verification includes the symmetrization argument and revised refinement parameter. The eight modules contain 256 checked theorems; the classical Markov inequality remains an explicit hypothesis. The individual axiom audits and aggregate source-hash record in the repository identify the checked development.

On October 20, 2026, thirty-one calculation, geometry, and abstract correctness theorems in `verification/lean/FPTASCalculations.lean` compiled successfully with Lean 4.34.0-rc2 and the pinned Mathlib revision. The checker `check_fptas_calculations.py` audits the transitive axioms of every theorem; the successful run is recorded in `fptas_calculations_verification_log.txt`. Only the standard axioms `propext`, `Classical.choice`, and `Quot.sound` occur, with no custom axioms or `sorry` placeholders. The verified scope is:

- *Slack geometry:* `interval_width` and `slack_extrapolation` prove the scalar width and nonnegativity inequalities used in Lemma 2.2, given the interval and affine-slack data. The declaration `linear_slack_progression` derives the slack identity from linearity of each constraint row; `reflecting_cell_progression` proves real feasibility in the chosen linear system from relative-width conditions, allowing constant-slack faces. Applying this to the shifted system and returning its integer samples to P is verified by the newer shifted theorems below. Thus feasibility of the extrapolated point is a conclusion, not an assumption. The finite construction of such cells and the full displacement-set inclusions are not formalized.

- *Integrality:* `mixed_integer_progression` proves that integer-spaced extrapolation preserves integrality of any specified subset of coordinates. The declaration `integer_slack_gap` proves that a nonnegative integer slack is either zero or at least one; it assumes the slack is integral and does not formalize matrix denominator clearing. These checks distinguish closure of the feasible domain under integer-spaced extrapolation from the discrete slack gap in the older unshifted construction, retained as an auxiliary result.

- *Error calculations:* `sampled_sup_algebra` proves $W \le R$ from $W \ge 0$, $c \le 1/2$, and $W \le R/2 + cW$. The declaration `second_difference_transfer` proves $p(1) - p(0) \le B$ from $p(-1) \ge p(0)$ and an assumed second-difference bound. These verify algebraic steps in Lemma 2.6, not its analytic estimates.

- *Refinement:* `refinement_error` proves $D^2/\ell^2 \le \varepsilon$ from $D \ge 0$, $\varepsilon, \ell > 0$, and $\ell \ge D/\sqrt{\varepsilon}$. For positive integer D, `ceiling_refinement` verifies positivity, $\ell \ge D^2$, and this error bound for the actual choice $\ell = \max\{D^2, \lceil D/\sqrt{\varepsilon} \rceil\}$.

- *Optimization transfer:* `local_comparison` verifies the scalar inequality in Proposition A.1, assuming selection, local optimization, and residual bounds. The declarations `uniform_model_error` and `nonnegative_maximization` verify, respectively, the $2\delta + \eta$ transfer for a uniformly accurate surrogate and the final algebra in Corollary 1.2.

The declarations `cover_solve_select` and `empty_candidates_iff` additionally verify the abstract selection and infeasibility arguments behind the procedure in Appendix A: a finite candidate set has a feasible minimizer with the stated error, and it is empty exactly when the feasible set is empty, assuming coverage and sound, complete local oracles. The theorem `representative_range_guarantee` verifies the range-relative selection step for arbitrary feasible representatives, assuming the error bound for one representative near an optimum. These results work over arbitrary point types, not just scalar decision variables. They establish conditional correctness implications, not implementations of the cover or local oracles. The later modules below verify the chamber cover and explicit encoding estimates. Enumeration, oracle implementations and the full running-time bound remain outside Lean. The earlier exploratory `ReflectingCells.lean` is not part of this successful check. The original `Unbounded-Cubic-Integer-Programming` repository separately records verification of 45 earlier cubic-integer-programming calculations; those results do not certify this FPTAS.

The module `verification/lean/SampledMarkov.lean` verifies thirteen results. The declarations `markov_difference`, `nearest_integer`, and `sampled_sup_bound` establish mean-value control, rounding, and the sample-to-supremum estimate. The declarations `coeff_reflect`, `degree_contract`, and `even_contract` construct the polynomial h from the even coefficients of q, prove $\deg h \le \lfloor D/2 \rfloor$, and prove $h(t^2) = (q(t) + q(-t))/2$. The theorem `symmetrized_difference` then proves the sharper factor $4\lfloor D/2 \rfloor^2/\ell^2$, and `sampled_markov` derives the stated D^2/ℓ^2 bound in Lemma 2.6. No symmetrization or approximation conclusion is assumed.

The proposition `MarkovBound` packages two explicit instances of the classical inequality: degree D on $[-\ell, \ell]$, with derivative factor D^2/ℓ, and degree $n = \lfloor D/2 \rfloor$ on $[0, \ell^2]$, with derivative factor $2n^2/\ell^2$. Neither is introduced as a Lean axiom. The retained helper `second_difference_bound` proves the former convexity-based estimate, but is no longer used by the strengthened sampled Markov proof. The declarations `representative_at_minimum`, `representative_at_maximum`, and `select_representative` propagate the new D^2/ℓ^2 error to extrema and finite selection, assuming feasible progression samples and global objective bounds. The checker `check_sampled_markov.py` records compilation and the thirteen-theorem axiom audit in `sampled_markov_verification_log.txt`.

The module `verification/lean/LineRestriction.lean` constructs $p(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ by substituting affine univariate polynomials into an actual `MvPolynomial`. Its first six checked theorems are `eval_restrict`, `eval_zero`, `eval_one`, `degree_restrict`, `multivariate_representative`, and `multivariate_selection`. They prove the evaluation identity, endpoint identities, and $\deg p \leq \deg f$ (including zero and constant restrictions), then compose these facts with the sampled Markov and selection theorems. Thus the final two theorems derive the approximation guarantee for multivariate polynomials from feasible progression samples, global objective bounds, and the classical Markov hypothesis; existence of a suitable univariate restriction is no longer assumed. The checker `check_line_restriction.py` rebuilds the sampled Markov dependency and records the fifteen-theorem axiom audit in `line_restriction_verification_log.txt`. All eight modules use only the standard axioms already listed and contain 256 checked results in total. The additional declaration `certified_cover_guarantee` combines finite coverage, sound and complete representatives, reflecting progressions, and the maximum/ceiling refinement into the final range-relative error bound. Its cover and progression properties are hypotheses. The additional theorems `slack_integral_progression` and `slack_cover_guarantee` define feasibility by linear inequalities and coordinate integrality, derive the progression samples from rowwise slack widths, and connect them to this final guarantee. The six additional results `integer_row_value`, `integer_half_shift`, `shifted_feasible_iff`, `shifted_slack_positive`, `shifted_slack_integral_progression`, and `shifted_slack_cover_guarantee` verify the new half-unit relaxation and connect closed shifted-slack intervals to the error guarantee on the original integer feasible set. CoverAssembly below supplies the chamber closures and their coverage; oracle implementations and full complexity remain outside the formalization.

Ten additional results in `FPTASCalculations.lean` verify the revised Markov-based grid argument. The declarations `ceil_weight_bounds` and `rounded_barycentric_weights` prove the actual ceiling-rounding inequalities, nonnegativity, and domination by $(1-t)\lambda_i$. The results `rounded_segment_weights` and `barycentric_coordinate_segment` construct residual convex weights and establish the segment identity coordinatewise; `barycentric_segment` gives the general weight decomposition. The theorem `rounded_grid_coordinate` proves membership in the denominator-$k\Delta$ grid via an explicit integer numerator. The theorems `markov_grid_multiplier` and `markov_grid_multiplier_bounds` verify positivity, the geometric threshold, the half-error bound, and the ceiling overhead. The results `markov_segment_error` and `markov_grid_transfer` verify the range-relative error budget without a Lipschitz constant or range-gap assumption. The additional result `unit_interval_markov_error` in `SampledMarkov.lean` derives the segment variation estimate from a classical interval Markov hypothesis using the mean value theorem. Integral-slice scaling and a Carathéodory representation with a largest weight remain external inputs. CoverAssembly below assembles the coordinatewise facts into the geometric reduction; the final bit complexity remains a human-review obligation. Earlier Lipschitz-based transfer helpers are retained but are not used by the revised paper proof. The new module `ScaledPolynomial.lean` constructs

the integer polynomial with coefficient $c_{\boldsymbol{\alpha}} Q^{D-\sum_{i\in I}\alpha_i}$, where I indexes the continuous coordinates. Its first eight theorems prove the exponent bound, coefficient formula, unchanged monomial support and total degree for $Q \neq 0$, and the evaluation identity $g(\mathbf{u}, \mathbf{z}) = Q^D f(\mathbf{u}/Q, \mathbf{z})$. The declaration `integer_representation` packages an actual integer-coefficient polynomial, its degree bound and the identity; these properties are conclusions, not hypotheses. Twelve further results verify inverse coordinate maps, the scaled-row identity, and a bijection between the scaled integer system and the compatible-grid system. They also prove the signed binary-length formula, the bound $\langle c'_{\boldsymbol{\alpha}} \rangle \leq \langle c_{\boldsymbol{\alpha}} \rangle + D\,\mathrm{size}(|Q|)$, and the corresponding summed coefficient-payload bound. Here size is binary magnitude length; Lean identifies the signed length with $1 + \mathrm{clog}_2(1 + |c|)$ using the natural ceiling logarithm. Six further results verify the scaled constraint lengths. Each matrix entry and right-hand side grows by at most $\mathrm{size}(|Q|)$ bits; the total dense constraint payload grows by at most $m(d+1)\mathrm{size}(|Q|)$. The same bound holds after adding unchanged dimension/list metadata. Twelve further results construct positive product denominators and integer entries for rational rows, including each right-hand side. They prove the real scaling identity and equivalence of the whole constraint system, with arbitrary additional domain restrictions. The denominator and entry lengths are bounded explicitly; a row with rational payload L has cleared integer payload at most $L(L+2)$. These bounds are also summed across rows. Ten further results check the mixed-integer complexity calculation. `continuous_basis_count` and `continuous_basis_power_bound` count candidate row bases. `integer_basis_determinant_size` bounds integer determinant lengths using the checked Leibniz estimate. `basis_lcm_spec` proves positivity and divisibility of the actual finite least common multiple, and `basis_lcm_size` bounds its length. `mixed_grid_size` propagates this to $Q = k\Delta$. The results `mixed_grid_polynomial_size` and `mixed_scaled_input_size` check polynomial encoding absorption. Finally, `mixed_runtime_composition` and `mixed_runtime_exponent` derive an explicit constant multiple of $d(d_c + 1)$ in the exponent from preprocessing and integer-solver bounds. These are conditional cost inequalities, not a verified implementation: normalization and grid-multiplier bit bounds, basis enumeration and arithmetic costs, the integer-solver cost bound, and assembly into the stated algorithm remain external or human-reviewed.

Verification and responsibility. Two further declarations, `unified_exponent_eq` and `unified_point_bounds`, express both bounds using the single exponent Φ; the sharper coordinate exponent remains an internal Lean helper. Nine preceding results verify the unshifted numerical estimates underlying Lemma 2.1; the additional half-unit in the upper slack bound is accounted for in the manuscript. The determinant expansion gives the factorial entry bound, and the nonzero integral denominator gives the Cramer coordinate bound. The declared exponents use the natural ceiling logarithm. Finite convex combinations preserve the coordinate bound, and the rowwise slack estimates give the strict dyadic upper bound. The combined declaration `represented_point_bounds` assumes Cramer data for vertices and a convex representation of the point; existence of these geometric data, the definition of τ via real logarithms, and the real/natural ceiling-log correspondence remain external inputs.

The module `verification/lean/CoverArithmetic.lean` adds 26 checked results. It translates the published CHKM coordinate and doubled-slack estimates, taken as hypotheses, to $\Phi = 5d^2(d+1)(\tau+3)+2$, including $d = 2$, and checks a separate one-dimensional estimate and convex-combination bounds. It constructs the finite threshold set using $2^k/4$ for $0 \leq k < \Phi + 2$, proves coverage of every relevant slack and the relative-width inequality, and bounds its cardinality. It verifies the binomial chamber recurrence, its power bound, and the ℓ^d oracle and ℓ^{d+1} LP-count bounds conditional on geometric insertion and per-stage cost hypotheses. The published CHKM estimates, the per-row encoding convention, existence of vertex representations, enumeration of chamber closures, and

algorithms and their bit costs remain external. These checks do not certify the complete chamber enumerator.

The module `verification/lean/CoverAssembly.lean` contains 47 checked results. It constructs generated interval cells and proves their cover and relative widths. It then proves coverage by closures of nonempty strict sign regions of the actual slack-threshold arrangement, including points on subdivision hyperplanes, and places each such closure in a narrow interval cell. The theorem `chamber_approximation` assembles this with the shifted-slack approximation guarantee. Proper rows, continuity, the upper slack bound, feasible complete representatives, extrema and Markov remain inputs. Separately, `mixed_integer_reduction` connects convex barycentric rounding, compatible-grid membership, the actual scaled integer polynomial and scaled feasible set, and the two error budgets. The structure `SliceData` records the published slice-scaling and Carathéodory data as explicit hypotheses; the scaled solver guarantee is also an input. The positive-parameter theorem treats positive continuous dimension and degree. Fourteen further results establish the rational descriptions: `closure_strict_linear` proves closure equality for a nonempty strict system; `chamber_closure_rational_description` supplies the actual rational matrix and right-hand side, including outer bounds. `chamber_closure_compressed` proves equality with the rational maximum lower and minimum upper slack bounds, giving two inequalities per row. `chamber_integer_tests` connects these to the rounded integer-slack tests. Nonemptiness of the strict chamber is essential. Ten additional results verify the common-margin LP criterion. `interior_linear_system` identifies the interior with strict feasibility for proper rows, and `strict_iff_positive_margin` constructs a common positive margin capped at one. `margin_feasible_iff` covers empty weak cells. `compressed_optimum_test` proves that a feasible optimum for the actual compressed rational system is positive exactly when the strict chamber is nonempty. Properness is derived from the original nonzero rows. Correctness and attainment of the LP optimum remain explicit inputs, not a verified solver. Enumeration, endpoint bit lengths, and subroutine implementations and costs remain outside this formalization.

The module `verification/lean/ChamberInsertion.lean` adds 21 checked results. It proves convexity of strict sign regions, constructs a crossing point for every split chamber, and injects split chambers into the induced sign regions. Translation to the kernel and rank-nullity prove the dimension reduction; constant restrictions are treated separately. The theorem `arrangement_count_bound` proves the binomial bound by induction without assuming the geometric recurrence. Reindexing, passage to closures, and convexity of the shifted-slack outer domain give `shifted_chamber_count` for the actual threshold family. No general-position assumption is needed. The earlier conditional recurrence lemma remains available, but geometric insertion is no longer a verification gap. The parallel-family refinement, enumeration implementation and subroutine bit costs remain outside this formalization.

The module `verification/lean/CellEncoding.lean` contains 39 checked results. It proves common-denominator representations for every actual threshold, reduced-denominator divisibility and binary-length bounds; compression preserves these bounds. The rational cell payload is at most $2m(d(\tau+2)+\tau+\Phi+4\,\mathrm{size}(\ell)+15)$. The rounded integer payload is at most $2m(d\tau+\tau+\Phi+2)$, including the original right-hand side and independent of ℓ. The encoded integer system is identified with chamber-closure membership. Common-margin LP coefficients and extra rows have checked overheads. The complete LP query, including its objective and an explicit dense metadata schema, is now assembled and bounded by $200(\varphi+1)^5$ under explicit input-parameter bounds. The actual refinement has checked bit length at most $2\,\mathrm{size}(D)+\mathrm{size}(q)$ for $\epsilon=a/q$. The normalized-entry, unary-degree and coefficient-log bridges are checked separately. A byte-level serializer, input parsing, and construction and oracle costs remain outside the formalization.

The assistant checked compilation of the edited LaTeX source and consulted literature during

development. At Hildebrand's request, Lean verification was resumed and the 256 results described above were checked; the complete manuscript is not formally verified. Successful typesetting does not verify a mathematical proof, and the literature searches do not establish novelty or exhaust the relevant literature. This disclosure does not assert that a complete independent human audit has already taken place. Responsibility for checking the arguments, complexity claims, citations, and contribution statements, and for approving any submitted version, remains with the human authors.